

SPL-1 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

Student's Name:	Md. Tofazzol Alam Rahat		
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Distance Vector Routing Simulator			
Project Description: The proposed project is a GUI-based, discrete-event Distance Vector Routing Simulator designed to demonstrate router communication, routing-table updates, and network changes through interactive visualization. The core routing and simulation backend will be implemented in C++, while the Java Swing interface will be used for GUI, visualization, and user interaction. The network topology will be created directly through the Java Swing interface, where users can add or remove routers and links and set or change link costs. The C++ backend will implement the Distance Vector routing algorithm using the Bellman-Ford algorithm without using any graph or routing library. It will support periodic and triggered updates, routing-table changes, link failure and restoration, convergence detection, routing loops, and the count-to-infinity problem. The simulator will provide different modes including Basic Distance Vector, Split Horizon, Poison Reverse, Hold-Down/Configurable Infinity, and Link State Routing using Dijkstra's algorithm. The GUI will also provide animated visualization of routing messages, where updates will be shown moving between routers and routing-table changes will be displayed dynamically during the simulation. The system will also provide controls for starting, pausing, resetting, and stepping through simulation rounds, along with a speed control and event/message log. The simulator will display routing tables, network changes, message activity, and convergence status for easier understanding and demonstration. If time permits and with supervisor approval, the simulator may additionally be developed and packaged as a standalone desktop application using Java JDK's "jpackage" tool for easier use and demonstration.			

Languages or Tools to be used:

- C++ programming language
- Java Swing
- Bellman-Ford algorithm
- Dijkstra's algorithm
- G++ compiler and Java JDK
- JSON / IPC for C++–Java communication
- Linux / WSL command line
- Git and GitHub for version control

No graph library, routing library, or network simulation framework will be used.

Supervisor's Name: Prof. Dr. Md. Shariful Islam

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback:

Before the Final Presentation:

I confirm that the project is complete and I am forwarding it for final presentation.

Signature of the supervisor: _____

Date: _____